

50.007 Machine Learning, Spring 2025

Homework 1

Due 28 February 2025, 11.59pm

Sample Solutions

In this homework, we would like to look at **Linear Classification** including perceptron algorithm and hinge loss, and **Regression** including linear regression and ridge regression.

1 Linear Classification [40 points]

1.1 Hinge Loss with Offset [15 points]

(1) Definition: Given a training set $\{(x_i, y_i)\}_{i=1}^n$, where $x_i \in \mathbb{R}^d$ and $y_i \in \{-1, +1\}$, the hinge loss function with an offset θ_0 is defined as

$$\ell(\theta, \theta_0; x_i, y_i) = \max\{1 - y_i(\theta \cdot x_i + \theta_0), 0\}.$$

The corresponding empirical risk is then given by

$$R_n(\theta, \theta_0) = \frac{1}{n} \sum_{i=1}^n \max\{1 - y_i(\theta \cdot x_i + \theta_0), 0\}.$$

(2) Stochastic Sub-Gradient Descent (SSGD) Procedure: To minimize the empirical risk $R_n(\theta, \theta_0)$ using SSGD, proceed as follows:

1. **Initialization:** Randomly initialize the parameters $\theta^{(0)}$ and $\theta_0^{(0)}$.
2. **Learning Rate:** For simplicity, use a fixed learning rate of $\eta = 0.1$ for all iterations. Another common strategy is to decrease it over time, such as $\eta_k = \frac{1}{k+1}$, to promote convergence.
3. **Iteration:** Repeat iteration until stopping criterion, e.g., maximum number of iterations or convergence of the empirical risk (when the change in the empirical risk between consecutive iterations is smaller than a set threshold), is met.

For each iteration $k = 0, 1, 2, \dots$, perform the following steps:

- (a) Randomly select an index $t \in \{1, 2, \dots, n\}$.
- (b) Compute the margin for the selected example:

$$m = y_t \left(\theta^{(k)} \cdot x_t + \theta_0^{(k)} \right).$$

(c) If $m > 1$, the hinge loss is zero and its sub-gradient vanishes. In this case, no update is made:

$$\theta^{(k+1)} = \theta^{(k)}, \quad \theta_0^{(k+1)} = \theta_0^{(k)}.$$

If $m \leq 1$, the sub-gradient is non-zero. The sub-gradients with respect to the parameters are given by

$$\nabla_{\theta} \ell(\theta, \theta_0; x_t, y_t) = -y_t x_t, \quad \nabla_{\theta_0} \ell(\theta, \theta_0; x_t, y_t) = -y_t.$$

Therefore, update the parameters as follows:

$$\theta^{(k+1)} = \theta^{(k)} + \eta_k y_t x_t,$$

$$\theta_0^{(k+1)} = \theta_0^{(k)} + \eta_k y_t.$$

4. **Best Parameter Tracking (Optional):** It may be helpful to keep track of the best parameters observed so far. That is, record the parameter set $(\theta^{(i)}, \theta_0^{(i)})$ that results in the lowest empirical risk

$$i^* = \arg \min_{i=1, \dots, k} R_n(\theta^{(i)}, \theta_0^{(i)}).$$

2 Linear Regression [40 points]

(1) [10 points] Given the training set

$$S_n = \{(x^{(t)}, y^{(t)}) \mid t = 1, 2, \dots, n\},$$

with $x^{(t)} \in \mathbb{R}^d$, the linear regression model with bias is

$$f(x; \theta, \theta_0) = \theta \cdot x + \theta_0.$$

The least squares cost function is

$$R_n(\theta, \theta_0) = \frac{1}{n} \sum_{t=1}^n \frac{1}{2} \left(y^{(t)} - (\theta \cdot x^{(t)} + \theta_0) \right)^2.$$

Closed-Form Solution: To incorporate the bias term θ_0 in a unified way with θ , we augment the input feature vector by adding a constant 1:

$$\tilde{x}^{(t)} = \begin{pmatrix} 1 \\ x^{(t)} \end{pmatrix} \in \mathbb{R}^{d+1},$$

and define the augmented parameter vector as

$$\tilde{\theta} = \begin{pmatrix} \theta_0 \\ \theta \end{pmatrix} \in \mathbb{R}^{d+1}.$$

Then, the model can be written as

$$f(x; \tilde{\theta}) = \tilde{\theta} \cdot \tilde{x}^{(t)},$$

and the cost function becomes

$$R_n(\tilde{\theta}) = \frac{1}{n} \sum_{t=1}^n \frac{1}{2} \left(y^{(t)} - \tilde{\theta} \cdot \tilde{x}^{(t)} \right)^2.$$

Now we use $\tilde{X} \in \mathbb{R}^{n \times (d+1)}$ denote the design matrix whose t -th row is $(\tilde{x}^{(t)})^T$, and let $\tilde{y} = [y^{(1)}, y^{(2)}, \dots, y^{(n)}]^T$. Define

$$\tilde{b} = \frac{1}{n} \tilde{X}^T \tilde{y}, \quad \tilde{A} = \frac{1}{n} \tilde{X}^T \tilde{X}.$$

Then the optimality condition can be written in compact form as

$$\tilde{A} \tilde{\theta} = \tilde{b}.$$

Assuming $\tilde{A}$ is invertible, the closed-form solution is given by

$$\tilde{\theta} = \tilde{A}^{-1} \tilde{b}.$$

Stochastic Gradient Descent with Offset: Using the augmented representation, the stochastic gradient descent (SGD) update rule becomes:

- **Initialization:** Randomly initialize $\tilde{\theta}^{(0)}$.
- **Iteration:** At each iteration k , randomly select an index $t \in \{1, 2, \dots, n\}$ and update:

$$\tilde{\theta}^{(k+1)} = \tilde{\theta}^{(k)} + \eta_k \left(y^{(t)} - \tilde{\theta}^{(k)} \cdot \tilde{x}^{(t)} \right) \tilde{x}^{(t)}.$$

Alternatively, the updates can be expressed separately for the bias and the weight vector:

$$\begin{aligned} \theta_0^{(k+1)} &= \theta_0^{(k)} + \eta_k \left(y^{(t)} - (\theta^{(k)} \cdot x^{(t)} + \theta_0^{(k)}) \right), \\ \theta^{(k+1)} &= \theta^{(k)} + \eta_k \left(y^{(t)} - (\theta^{(k)} \cdot x^{(t)} + \theta_0^{(k)}) \right) x^{(t)}. \end{aligned}$$